

FEATURES

- Synchronous Parallel Load
- Positive-Edge-Triggered Clocking
- J and $\bar{K}$ Inputs to First Stage
- Complementary Outputs From Last Stage
- Package Options: Plastic and Ceramic DIPS and Ceramic Chip Carriers
- Dependable Texas Instruments Quality and Reliability

DESCRIPTION/ORDERING INFORMATION

These 4-bit registers feature parallel inputs, parallel outputs, J- $\bar{K}$ serial inputs, shift/load control input, and a direct overriding clear. The registers have two modes of operation: parallel (broadside) load, and shift (in the direction Q_A and Q_D).

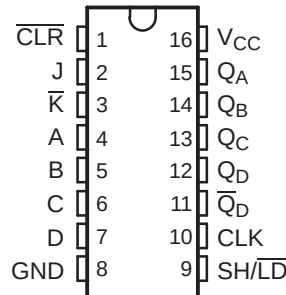
Parallel loading is accomplished by applying the 4-bits of data and taking the shift/load control input low. The data is loaded into the associated flip-flop and appears at the outputs after the positive transition of the clock input. During loading, serial data flow is inhibited.

Shifting is accomplished synchronously when the shift/load control input is high. Serial data for this mode is entered at the J- $\bar{K}$ inputs. These inputs permit the first stage to perform as a J- $\bar{K}$, D, or T type flip-flop as shown in the function table.

The SN54HC195 is characterized for operation over the full military temperature range of -55°C to 125°C .

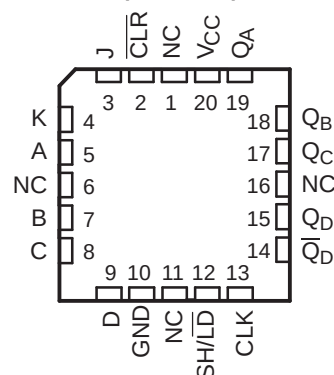
SN54HC195 . . . J PACKAGE

(TOP VIEW)



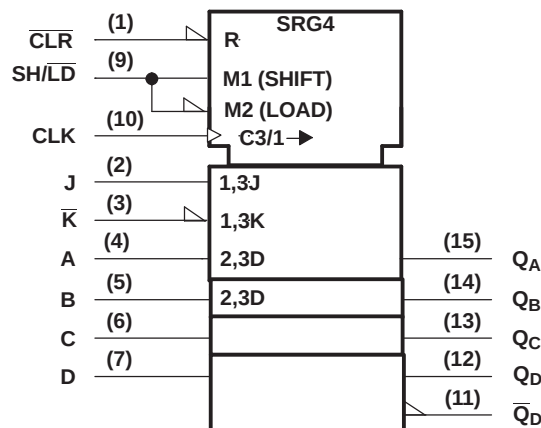
SN54HC195 . . . FK PACKAGE

(TOP VIEW)



NC – No internal connection

LOGIC SYMBOL[†]



[†] This symbol is in accordance with ANSI/IEEE Std 91-1984 and IEC Publication 617-12.

Pin numbers shown are for J package.



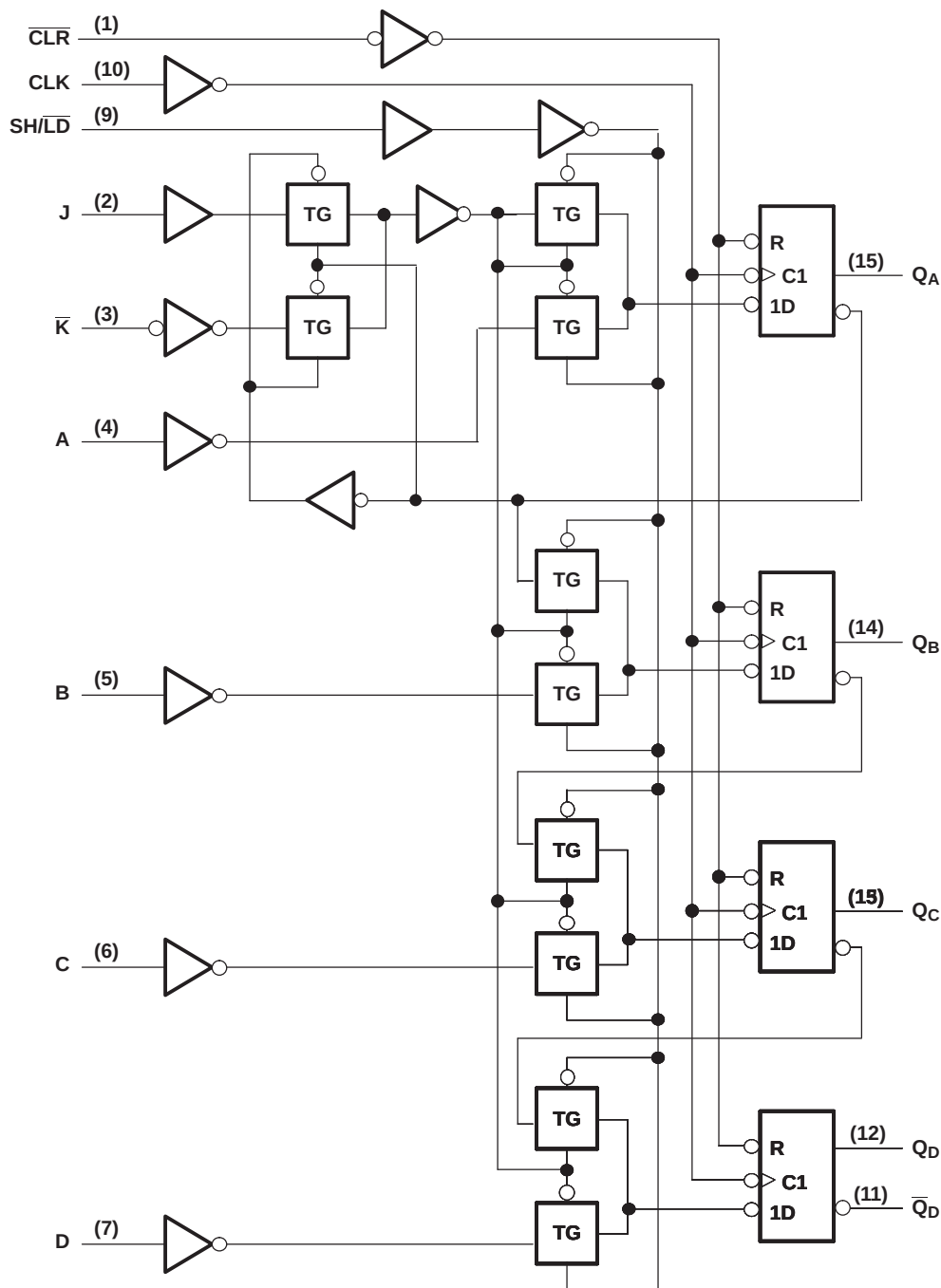
Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

SN54HC195

4-BIT PARALLEL-ACCESS SHIFT REGISTERS

SCLS124A–DECEMBER 1992–REVISED NOVEMBER 2007

LOGIC DIAGRAM (POSITIVE LOGIC)

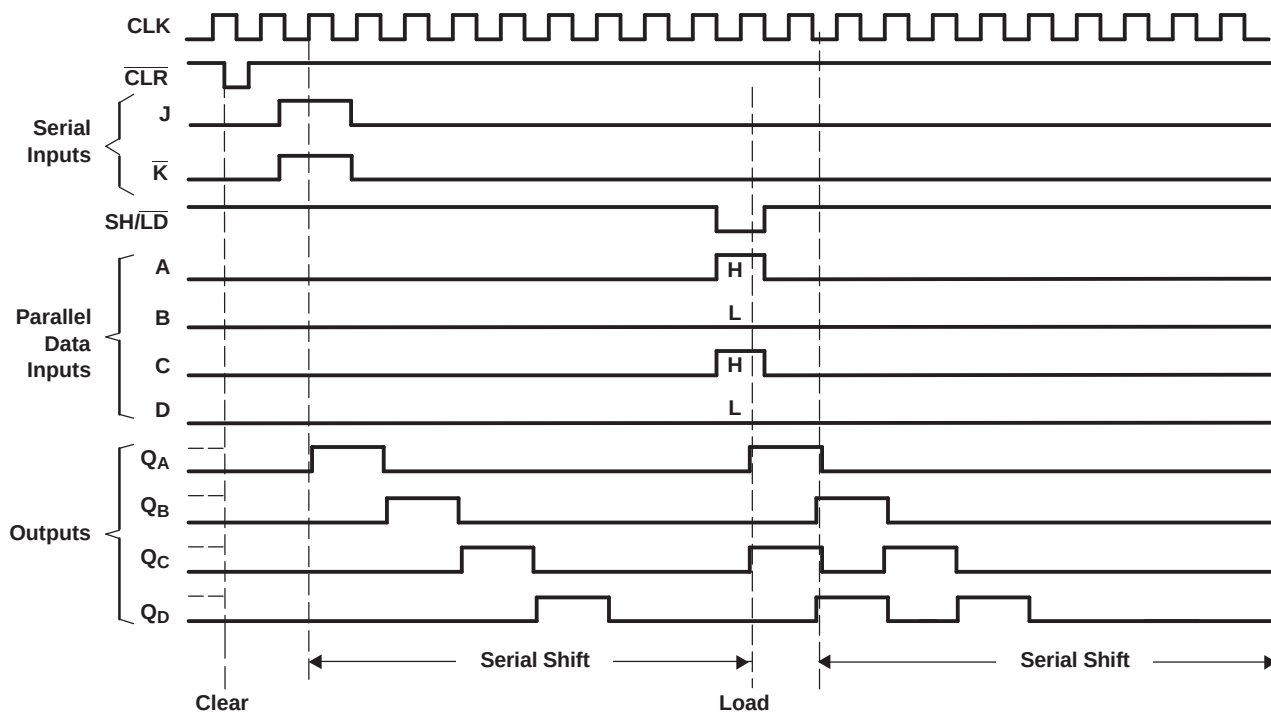


Pin numbers shown are for J package.

FUNCTION TABLE

CLR	SH/LD	CLK	INPUTS						OUTPUTS				
			SERIAL		PARALLEL				Q _A	Q _B	Q _C	Q _D	$\overline{Q}_D$
			J	$\overline{K}$	A	B	C	D					
L	X	X	X	X	X	X	X	X	L	L	L	L	H
H	L	↑	X	X	a	b	c	d	a	b	c	d	$\overline{d}$
H	H	L	X	X	X	X	X	X	Q _{A0}	Q _{B0}	Q _{C0}	Q _{D0}	$\overline{Q}_{D0}$
H	H	↑	L	H	X	X	X	X	Q _{A0}	Q _{A0}	Q _{Bn}	Q _{Cn}	$\overline{Q}_{Cn}$
H	H	↑	L	L	X	X	X	X	L	Q _{An}	Q _{Bn}	Q _{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	H	X	X	X	X	H	Q _{An}	Q _{Bn}	Q _{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	L	X	X	X	X	$\overline{Q}_{An}$	Q _{An}	Q _{Bn}	Q _{Cn}	$\overline{Q}_{Cn}$

TYPICAL CLEAR, SHIFT, AND LOAD SEQUENCES



SN54HC195

4-BIT PARALLEL-ACCESS SHIFT REGISTERS

SCLS124A–DECEMBER 1992–REVISED NOVEMBER 2007

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V_{CC}	Supply voltage range	–0.5	7	V
I_{IK}	Input clamp current $V_I < 0$ or $V_I > V_{CC}$		±20	mA
I_{OK}	Output clamp current $V_O < 0$ or $V_O > V_{CC}$		±20	mA
I_O	Continuous output current $V_O = 0$ to V_{CC}		25	mA
	Continuous current through V_{CC} or GND pins		50	mA
	Lead temperature 1,6 mm (1/16 in) from case for 60 s: FK or J package		300	°C
	Lead temperature 1,6 mm (1/16 in) from case for 10 s: N package		260	°C
T_{stg}	Storage temperature range	–65	150	°C

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

RECOMMENDED OPERATING CONDITIONS

		MIN	NOM	MAX	UNIT
V_{CC}	Supply voltage	2	5	6	V
V_{IH}	High-level input voltage	$V_{CC} = 2$ V	1.5		V
		$V_{CC} = 4.5$ V	3.15		
		$V_{CC} = 6$ V	4.2		
V_{IL}	Low-level input voltage	$V_{CC} = 2$ V	0	0.3	V
		$V_{CC} = 4.5$ V	0	0.9	
		$V_{CC} = 6$ V	0	1.2	
V_I	Input voltage	0		V_{CC}	V
V_O	Output voltage	0		V_{CC}	V
t_t	Input transition (rise and fall) times	$V_{CC} = 2$ V	0	1000	ns
		$V_{CC} = 4.5$ V	0	500	
		$V_{CC} = 6$ V	0	400	
T_A	Operating free-air temperature	–55		125	°C

ELECTRICAL CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS	V_{CC}	$T_A = 25^\circ\text{C}$			SN54HC195		UNIT
			MIN	TYP	MAX	MIN	MAX	
V_{OH}	$V_I = V_{IH}$ or V_{IL} , $I_{OH} = -20\ \mu\text{A}$	2 V	1.9	1.998		1.9		V
		4.5 V	4.4	4.499		4.4		
		6 V	5.9	5.999		5.9		
	$V_I = V_{IH}$ or V_{IL} , $I_{OH} = -4\ \text{mA}$	4.5 V	3.98	4.30		3.7		
		6 V	5.48	5.80		5.2		
V_{OL}	$V_I = V_{IH}$ or V_{IL} , $I_{OL} = 20\ \mu\text{A}$	2 V		0.002	0.1		0.1	V
		4.5 V		0.001	0.1		0.1	
		6 V		0.001	0.1		0.1	
	$V_I = V_{IH}$ or V_{IL} , $I_{OL} = 4\ \text{mA}$	4.5 V		0.17	0.26		0.4	
		6 V		0.15	0.26		0.4	
I_I	$V_I = V_{CC}$ or 0	6 V		±0.1	±100		±1000	nA
I_{CC}	$V_I = V_{CC}$ or 0, $I_O = 0$	6 V			8		160	μA
C_I	$V_I = V_{CC}$ or GND	2 V to 6 V		3	10		10	pF

TIMING REQUIREMENTS

over recommended operating free-air temperature range (unless otherwise noted)

			V _{CC}	T _A = 25°C		SN54HC195		UNIT
				MIN	MAX	MIN	MAX	
f _{clock}	Clock frequency		2 V	0	6	0	4.2	MHz
			4.5 V	0	31	0	21	
			6 V	0	36	0	25	
t _w	Pulse duration	CLK high or low	2 V	80		120		ns
			4.5 V	16		24		
			6 V	14		20		
	$\overline{\text{CLR}}$ low	2 V	80		120			
		4.5 V	16		24			
		6 V	14		20			
t _{su}	Setup time, before CLK↑	SH/ $\overline{\text{LD}}$, or serial and parallel data, or CLR inactive	2 V	100		150		ns
			4.5 V	20		30		
			6 V	17		26		
t _h	Hold time, after CLK↑	SH/ $\overline{\text{LD}}$, or serial and parallel data, or CLR inactive	2 V	0		0		ns
			4.5 V	0		0		
			6 V	0		0		

SWITCHING CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted), $C_L = 50 \text{ pF}^{(1)}$

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V _{CC}	T _A = 25°C			SN54HC195		UNIT
				MIN	TYP	MAX	MIN	MAX	
f _{max}			2 V	6	12		4.2		MHz
			4.5 V	31	50		21		
			6 V	36	60		25		
t _{pd}	CLK	Q _A thru Q _D or Q _D	2 V		67	145		220	ns
			4.5 V		17	29		44	
			6 V		14	25		37	
t _{pd}	CLR	Q _A thru Q _D or Q _D	2 V		67	150		225	ns
			4.5 V		17	30		45	
			6 V		13	26		38	
t _t		Any	2 V		28	75		110	ns
			4.5 V		8	15		22	
			6 V		6	13		19	
C _{pd}	Power dissipation capacitance			No load, T _A = 25°C					65 pF typ

(1) Load circuit and voltage waveforms are shown in previous pages.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-8682701EA	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8682701EA SNJ54HC195J
SN54HC195J	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	SN54HC195J
SN54HC195J.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	SN54HC195J
SNJ54HC195J	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8682701EA SNJ54HC195J
SNJ54HC195J.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8682701EA SNJ54HC195J

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

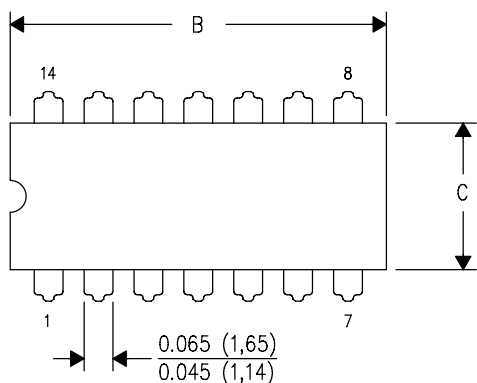
Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

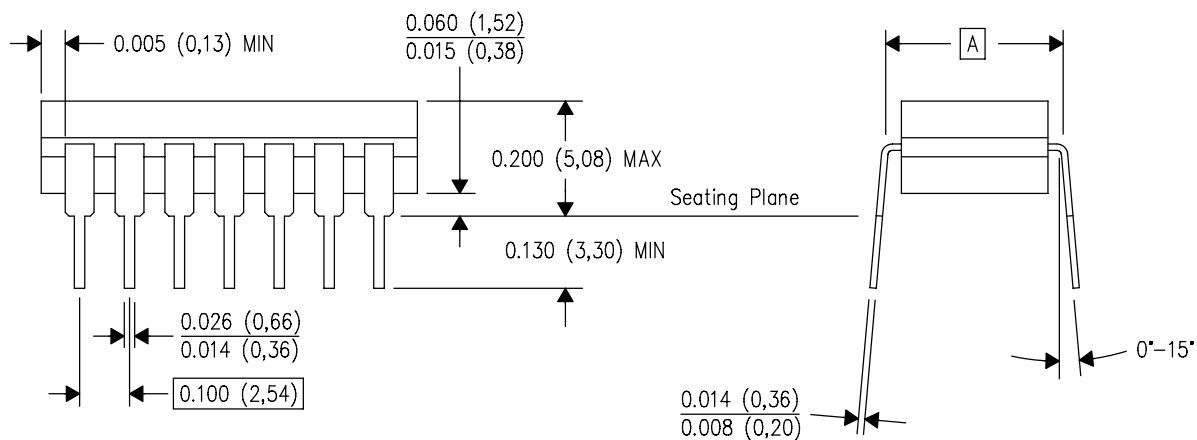
J (R-GDIP-T**)

14 LEADS SHOWN

CERAMIC DUAL IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)



4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. This package is hermetically sealed with a ceramic lid using glass frit.
 - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
 - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.

IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATASHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES "AS IS" AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, regulatory or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you fully indemnify TI and its representatives against any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI's products are provided subject to [TI's Terms of Sale](#), [TI's General Quality Guidelines](#), or other applicable terms available either on [ti.com](#) or provided in conjunction with such TI products. TI's provision of these resources does not expand or otherwise alter TI's applicable warranties or warranty disclaimers for TI products. Unless TI explicitly designates a product as custom or customer-specified, TI products are standard, catalog, general purpose devices.

TI objects to and rejects any additional or different terms you may propose.

Copyright © 2025, Texas Instruments Incorporated

Last updated 10/2025